

BCE-O633
SEMESTER EXAMINATION-2023
CLASS – B.TECH SEMESTER VI
INTRODUCTION TO DATA SCIENCE AND DESIGN THINKING

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any five questions in about 150 words each. Each question carries six marks. (5 X 6 = 30 Marks)

Question-1: Explain the term Data Science and discuss its application.

Question-2: Discuss the Data Science Roles in development of Data Science projects.

Question-3: What are the 5 properties of data with respect to Data Science.

Question-4: Define Data transformation and its challenges.

Question-5: Define Data Pre -Processing? Why it is required?

Question-6: Elaborate the process of Data Reduction.

Question-7: Discuss the concept of Empathy, Sympathy and communication in design thinking

Question-8: Discuss the properties of a good design. What are the differences of good design and bad design?

Question-9: What do you mean by Heat map? Where it is used.

Question-10: How to incorporate the design thinking principles in data science? Discuss

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any FOUR questions in detail. Each question carries 10 marks. (4 X 10 = 40 Marks)

Question-1: What are the Data Security Issues? Write any 5 data security issues in detail.

Question-2: What is Data Collection? What are the common methods of data collection?

Question-3: What are the stages in a data science project and how it maps with data analytics life cycle

Question-4: Find mean, median and mode of the following data and then create a box plot.

24, 22, 34, 112, 12, 145, 56, 78, 90, 34, 21, 67, 90, 55, 23, 50, 34

Question-5: What is Data Integration? What are the challenges of data integration?

Question-6: Discuss the stages of design thinking process? Explain each stage in detail with example.

Question-7: Discuss various applications of Design thinking for Rural Development.

Question-8: What is the importance of Nature inspired designing? Discuss the examples of design that are inspired from nature.